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324631 (24)

BE (6th Semester)

Examination, April-May, 2014

Branch : Electrical

FIBRE OPTICS

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : (i) Part (a) of every question is compulsory.

(ii) Answer any two parts from b, c and d parts of each question.

Q. 1. (a) Define acceptance angle and numerical aperture of an optical fibre. 2

(b) Explain different modes in optical fibres and their configuration. 7

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(c) Describe briefly the salient features of vapour axial deposition (VAD) and the plasma activated chemical vapour deposition (PCVD) of fibre fabrication techniques. 7

(d) A graded index fibre with a parabolic index profile supports the propagation of 742 guided modes. The fibre has a numerical aperture in air of 0.3 and a core diameter of 70 μm. Determine the wavelength of the light propagation in the fibre. Also estimate the maximum diameter of the fibre which gives single mode operation at the same wavelength. 7

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Q. 2. (a) Define "external" and "internal" quantum efficiencies of an LED. 2

(b) What do you mean by "population inversion" ? Explain how it can be achieved in case of two energy level, three energy level and four energy level systems. 7

(c) Discuss the mechanism of optical feedback to provide oscillation and hence amplification within the laser. Indicate how this provides a distinctive spectral output from the device. 7

(d) The quantum efficiency of a particular silicon RAPD is 80% for the deflection of radiation at

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a wavelength of 0.9 μm. When the incident optical power is 0.5 μW, the output current from the device (after avalanche gain) is 11 μA. Determine the multiplication factor of the photo diode under these conditions. 7

Q. 3. (a) What do you mean by signal degradation in optical fibre ? What are its reason ? 2

(b) Discuss various types of dispersion in optical fibres. 7

(c) Explain the linear scattering losses and non-linear scattering losses in brief. 7

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(d) A multimode step index fibre has a numerical aperture of 0.3 and a core refractive index of 1.45. The material dispersion parameter for the fibre is 250 psn km^{-1} which makes material dispersion the totally dominating intramodal dispersion mechanism. Estimate :

(i) the total rms pulse broadening per kilometer, when the fibre is used with an LED source of rms spectral width 50 nm.

(ii) the corresponding bandwidth-length product for the fibre. 7

Q. 4. (a) What are the key system requirements needed in the analysis and design of an optical link ? 2

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(b) Explain RZ and NRZ format used for optical fibre transmission links, with its time domain signal and power spectrum. 7

(c) Calculate the number of received photons per bit in order to maintain a BER of 10^{-9} for :

(i) Ask heterodyne synchronous detection

(ii) Ask heterodyne non-synchronous detection. An ideal binary receiver may be assumed in all cases. 7

(d) Explain link power budget and rise-time budget. 7

Q. 5. (a) Name the different topologies used for fibre optic networks. 2

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(b) Explain unidirectional and bidirectional WDM and the system requirements for WDM. 7

(c) Draw a block diagram of SONET architecture and describe its operation. 7

(d) Give a note on optical amplifiers. 7

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